

1.1 Modal Analysis User Guide: Stochastic Subspace Identification (SSI)

The covariance-driven stochastic subspace identification (Cov-SSI) method is one of the most powerful techniques for operational modal analysis. This method is implemented in the Sensequake Cloud platform and can be accessed from the **Analysis** menu within a project.

The first step is to upload sensor data via the **Project Data** menu. The platform supports multi-setup surveys, where one or more sensors are used as references. Users can upload Sensequake Larzé data files in binary format or CSV files, with sensor data arranged in columns.

The Cov-SSI method is divided into three steps: preprocessing, singular value decomposition (SVD), and modal analysis. These steps are usually performed in sequence, but users can also execute all three steps at once if configuration files are already prepared.

1.1.2 Preprocessing

The preprocessing step allows users to select the data source, apply basic signal processing, and configure preliminary parameters for Cov-SSI. The subsequent steps will use this preprocessed data for the SSI calculations. This step includes several parameter sections:

1.1.2.1 Main Settings

There are two parameters in the **Main Settings** section.

The first parameter, **Time Lag**, defines the data length used for SSI estimation. The number of data points used to build the Hankel matrix in SSI is calculated as:

$$\text{Number of data points} = \text{Time lag (s)} \times \text{Sampling frequency.}$$

Users should note that the **Decimation** option in the **Preprocessing** → **Signal Processing** section affects the sampling frequency. The adjusted sampling frequency is equal to the original sampling frequency divided by the decimation factor.

The second parameter, **Number of Singular Values**, determines how many singular values from the Frequency Domain Decomposition (FDD) are plotted in the final results.. This plot helps users identify the physical poles in the SSI analysis during later steps. See also <https://sensequake.atlassian.net/l/cp/fH1Mtme5>

Can't find link.

1.1.2.2 Data Channels

The **Data Channels** section allows users to select specific channels for SSI analysis. Three parameters can be adjusted in this section: **Channels**, **Removed Sensor IDs**, and **Reference Sensors**.

The **Channels** parameter offers six options: **Vx**, **Vy**, **Vz**, **Ax**, **Ay**, and **Az**. “V” represents velocity data, while “A” represents acceleration data. “X,” “Y,” and “Z” correspond to the measurement axes. It is not recommended to mix velocity and acceleration channels. In cases where motion is dominant in one axis while others are relatively inactive, selecting only the active axis and muting the others may improve the quality of the identified modes.

The **Removed Sensor IDs** option allows users to mute data from specific sensors, which is useful when a sensor malfunctions.

The **Reference Sensor** option lets users select one or more sensors to serve as references during SSI. Using reference sensors reduces computational effort and thereby accelerates SSI computation. Note that the reference sensor in this setting is not necessarily the same as the reference nodes used in multi-setup tests. Users should choose sensors that provide high-quality signals across the frequency spectrum; otherwise, the quality of results (natural frequencies, mode shapes, etc.) may degrade significantly [1].

1.1.2.3 Signal Processing

The **Signal Processing** section allows users to modify data extracted from the sensors. It offers four optional data processing methods: filtering, equalization, decimation, and cleanup.

The **Filter** option supports four types of filters: lowpass, highpass, bandpass, and bandstop. Users can specify the order of each filter arbitrarily. The highpass and lowpass filters require one critical frequency, while the bandpass and bandstop filters require two. Multiple filters can be applied simultaneously. For more details, users can refer to [SciPy v1.16.0 Manual](#).

Equalization is a filter that balances the geophone signal, which tends to decay below 4 Hz. This option can improve the quality of velocity signals in the low-frequency range and facilitate peak picking.

The **Decimation** option downsamples the signal after applying an anti-aliasing filter. It can speed up computation when high-frequency components are less critical. Users may refer to [SciPy v1.16.0 Manual](#).

The **Cleanup** option trims portions of the signal that exceed the sensor's operational limits. This is useful for mitigating unexpected interferences, such as sudden impacts or sensor displacements that may occur during data collection. The cleanup function automatically filters out data segments exhibiting outlier levels of acceleration or velocity. This process is applied uniformly across all sensors, ensuring seamless synchronization.

1.1.3 SVD computation

The SVD computation is straightforward: select the target setup or setup group, then click the **Submit** button.

1.1.4 Modal Analysis

The **Modal Analysis** section computes the system's modes by postprocessing the SVD results. This postprocessing involves two main components: the **stability diagram** and **automated clustering**. Parameters for both components can be configured in this section.

The **Lowest Modal Order** and **Highest Modal Order** define the range of model orders used to generate the stability diagram. The model order indicates the number of largest singular values used for modal estimation. At different orders, the identified modes (poles) may exhibit varying behaviors: a true physical mode typically appears with stable frequency and damping values across multiple orders, while noise modes tend to be unstable and inconsistent. A semi-automated clustering algorithm is applied to help distinguish physical modes from noise.

The semi-automated clustering method is based on Reynders' fully automated Operational Modal Analysis (OMA) approach [2]. In Reynders' method, physical modes are first distinguished from noise modes using several pole parameters. Then, hierarchical clustering is applied to group similar modes, using a threshold defined as **mean + 2 × standard deviation** of the selected parameters.

The modified clustering method used here follows the same principle for separating physical and noise modes, but allows the use of custom parameters to identify noisy poles. In this method, the hierarchical clustering threshold is defined as:

Clustering Threshold = Clustering Threshold Factor × mean.

A larger **Clustering Threshold Factor** results in fewer, larger clusters of modes, while a smaller factor produces more clusters of smaller size.

1. Peeters, Bart, and Guido De Roeck. "Reference-based stochastic subspace identification for output-only modal analysis." *Mechanical systems and signal processing* 13.6 (1999): 855-878.
2. Reynders, Edwin, Jeroen Houbrechts, and Guido De Roeck. "Fully automated (operational) modal analysis." *Mechanical systems and signal processing* 29 (2012): 228-250.

1.2 Modal Analysis: Getting Started

This page is intended to help users learn how to use the Dashboard through hands-on experience. A one-story building example and a three-story building example will be presented, and the modal analysis of each structure using SSI will be demonstrated step by step.

Example: One-Story Building

We begin with a **one-story building** example. A 3D view of the structure is shown in Figure 1. The floor dimensions are 25 m by 31 m, and the story height is 8 m. Figure 1 also illustrates the global Cartesian coordinate system, node numbering, and overall geometry.

Note: the origin of the coordinate system is located at Node #1.

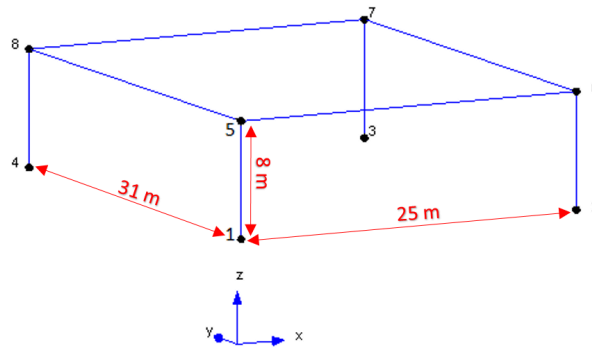


Figure 1: 3D View of the One-Story Building Model

Setting Up the Project

First, create a new project by clicking “+ NEW PROJECT” in the top-right corner of the homepage. From the dropdown menu, select “Larzé Ambient Vibration System.” This will take you to the project setup page.

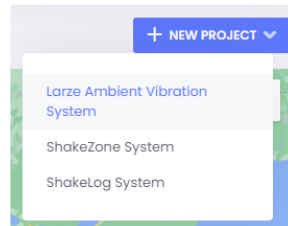


Figure 2: Creating a Larzé Project

The project setup page includes several sections: **Project Details**, **HAZUS Damage Estimation**, **Location**, **Time Zone**, **Units**, **Geometry**, and **Notes**. To begin, users can name the project in the **Project Details** section, then focus on the **Geometry** section for this example.

Within the **Geometry** section, multiple steps are required to define the structure's geometry. Information can be entered manually or uploaded via tab-delimited text files at each step.

The first step is to define the nodes. The **Nodes** section allows users to specify each node's ID, X, Y, Z coordinates, Sensor ID, and Setup ID. For the one-story building example, the **Nodes** section is shown in Figure 3.

Change Shape ×

ID	X	Y	Z	Sensor ID	Setup ID (0 for Reference Node)	+
1	0	0	0			 
2	25	0	0			 
3	25	31	0			 
4	0	31	0			 
5	0	0	8	5	1	 
6	25	0	8	6	1	 
7	25	31	8	7	1	 
8	0	31	8	8	1	 

Choose File No file chosen

Figure 3: Geometry — Nodes Settings

Next, define the lines. Each line connects two nodes. The line settings for the project are shown in Figure 4.

Change Shape ×

Node_1	Node_2	+
1	5	 
2	6	 
3	7	 
4	8	 
5	6	 
6	7	 
7	8	 
8	5	 

Choose File No file chosen

Figure 4: Geometry — Lines Settings

Next, define the surfaces. The current version of the Dashboard supports only triangular meshes; quadrilateral meshes are not supported at this time. The surface settings for this example are shown in Figure 5.

Change Shape ×

Node_1	Node_2	Node_3	+
5	6	7	 
5	7	8	 

Choose File No file chosen

Figure 5: Geometry — Surfaces Settings

The remaining settings are related to HAZUS estimation and are not covered in this tutorial. Click the “✓” (**Update**) button to save your changes. Users can verify the geometry by clicking “**Preview**”, which displays the 3D model of the example (see Figure 6). Once satisfied, click the “**Submit**” button at the bottom of the page to send the changes to the server.

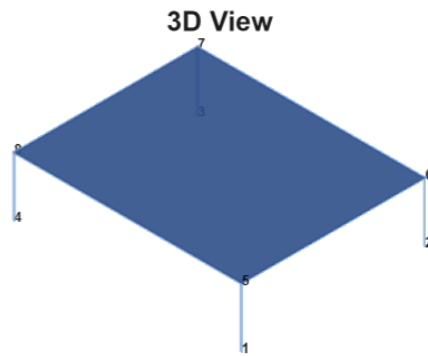


Figure 6: Geometry — 3D Structure Preview

The **Geometry** section also allows users to configure **Setup Info**, which refers to multi-setup tests. This part is skipped in the current example since only a single setup is involved. It will be discussed in more detail in the three-story building example.

Uploading the Data

Users can upload **.bin** files recorded from Larzé sensors. The Larzé Dashboard automatically detects the setup ID and sensor ID from the filename, so the filename should only be modified when necessary. The Dashboard also automatically reads the serial number, start time, duration, sampling frequency, and signal channels. The project data interface is shown in Figure 7. Note that the **Start Time** field may vary depending on the user's time zone.

Project Data

Setup Number:

Setup 1

Upload

Upload Non-Sensequake Data

Download

Delete

☐ Sensor ID	Serial Number	Start Time	Duration (min)	Sampling Frequency (Hz)	Channels
☐ 5		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx
☐ 6		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx
☐ 7		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx
☐ 8		Fri Dec 31 1999 19:00:00 GMT-0500 (Eastern Standard Time)	7	128	Vx

Figure 7: Project Data for One-Story Building

Once data is uploaded, the **Setup Info** section in **Project Settings** → **Geometry** will update automatically. Users can assign a **Setup Group** number to each setup to facilitate processing. This step is skipped in the current example since only one setup exists.

SSI Modal Analysis

As outlined in the documentation, the SSI process consists of three steps: preprocessing, SVD computation, and modal analysis. These steps will be introduced one by one in the following section. Users may follow the step-by-step process or run the analysis directly by uploading a configuration file.

Preprocessing

First, select **“Setup 1”** from the **Setup Number** menu. In the **Main Settings**, set the **Timelag** to 0.5 seconds. Since the data is recorded at a sampling frequency of 128 Hz, 0.5 seconds corresponds to 64 timelag points for SSI. A timelag of 64 is appropriate for this simple case, as higher values may introduce more noise and increase computation time. Set the **Number of Singular Values** to 1.

In the **Data Channels** section, select **“Vx”** since it is the only available data channel. No sensors are removed, and no reference sensors are used. Additionally, no signal processing is required for this example.

Once all parameters are set, click the **“Submit”** button, as shown in Figure 8.

Setup Number

Setup 1

Main Settings

Timelag (s)

0.5

Number of Singular Values to plot

1

Data Channels

Channels

☒ Vx

☐ Vy

☐ Vz

☐ Ax

☐ Ay

☐ Az

Removed Sensor IDs

☐ Use reference sensor(s)

Signal Processing

☐ Filter

Use Equalization

Decimate

Perform Cleanup

Figure 8: Preprocessing Parameters for One-Story Building Example

SVD Computation

Click the **“Submit”** button to initiate the server-side SVD computation for SSI.

Modal Analysis

In Modal Analysis, the model order parameters range from 2 to 50. The **Clustering Threshold Factor** is set to 0.5. For the initial analysis, these are considered “safe” default values, which can be adjusted based on the results of the first trial.

Setup Number

Setup 1

Lowest Model Order

2

Highest Model Order

50

Clustering Threshold Factor

0.5

Figure 9: Modal Analysis Parameters for One-Story Building Example

Results

Once all three steps of SSI are complete, the results page is displayed. Clicking **“Load Result”** on the Modal Analysis page will open the SSI results page. The SSI results for the one-story building, using the parameters described above, are shown in Figures 10, 11, 12, and 13.

The results page includes three plots:

- The left plot, titled **“Modal Analysis after Filtering”**, displays the SSI poles, with good poles shown in green and bad poles in red.
- The middle plot shows the frequencies and damping ratios of the clustered modes.
- The right plot displays the size of each cluster.

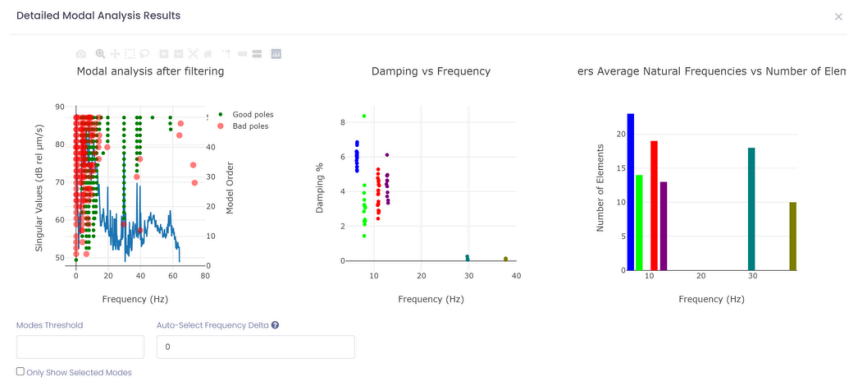


Figure 10: SSI Results of One-Story Building, Part I



Figure 11: SSI Results of One-Story Building, Part II

Figure 11 presents detailed information on each cluster. The table on the left displays key data for each cluster, including cluster size (number of poles), frequency (mean and standard deviation), and damping (mean and standard deviation).

The last two figures illustrate detailed mode visualization tools. Figure 12 shows the polar chart of the mode, which is used to visualize the mode phase. Since modes are complex values, the phase is essential for distinguishing physical modes from noise. Figure 13 presents the animation of the mode shape.

In this demo, we focus on the mode at 6.407 Hz as an example. According to the Figure 11 table, this mode forms a cluster with 23 poles. The cluster's average frequency is 6.407 Hz, with a standard deviation of 0.037 Hz. The average damping ratio is 6.052%, with a standard deviation of 0.436%. The polar chart in Figure 12 demonstrates that the nodal values align closely in phase, clustering near 0°, indicating a stable physical mode.

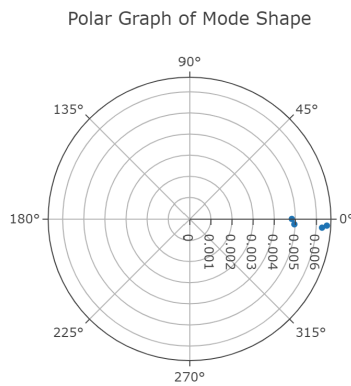


Figure 12: Polar graph of the mode at 6.407 Hz

Figure 13 shows that the mode is an in-plane transversal mode in the x-direction. The toolbar at the top allows users to control the plot perspective with five options: four-view plot, front view, top view, side view, and 3D view. The animation can be paused and saved as a GIF file..

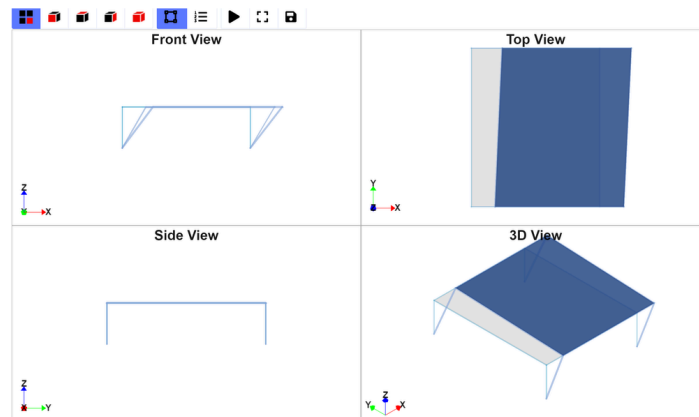


Figure 13: Mode shape of the mode at 6.407 Hz

Example: Three-Story Building

The three-story building is the second example demonstrated in this document. This example uses a reference sensor and involves multiple tests. The demo will focus on the multi-setup test analysis. The geometry of the three-story building is shown in Figure 14.

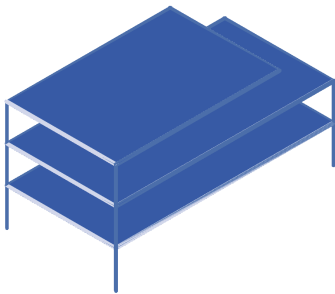


Figure 14: Geometry of Three-Story Building

Setting Up the Project

The steps to create the project are similar to those for the one-story building.

After creating the project, users can begin working in the **Geometry** tab for the three-story building. Defining the geometry involves a few additional steps, including specifying the setup and sensor IDs assigned to each sensor. In this example, the layout is predefined, and the mode shapes are known. In real-world applications, users should follow these guidelines for layout design:

- The reference sensor must remain the same across different setups.
- Only non-reference sensors may be moved between setups.
- Reference sensors should be placed at locations that are well excited by different modes.

Note: The steps for defining lines and surfaces are the same as those described for the one-story building example.

Uploading the Data

The **.bin** files should be uploaded to the Dashboard using the same method as before. The Dashboard will automatically organize the setups. As shown in Figure 15, the Dashboard recognizes two setups based on the filenames.

Once the data files are uploaded, the **Setup Info** section in **Project Settings** will update automatically. To analyze Setup 1 and Setup 2 together, set the **Setup Group** of both setups to the same value.

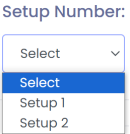


Figure 15: Setup Numbers

Setup Info				
Setup ID	Setup Group (0 for none)	Structure Name	Geometry Name	
1	1		INITIAL	
2	1		INITIAL	

Figure 16: Setup Info Section in Project Settings

SSI Modal Analysis

In this section, SSI is performed using the single-step analysis option. While the single-step method is more convenient, the three-step process offers greater flexibility, as each stage can be fine-tuned individually.

In this example, Setup 1 and Setup 2 are analyzed simultaneously as **Group 1**, using the same parameters. However, they can also be analyzed separately with different parameter sets if needed. The parameters used for Group 1 are similar to those in the previous example. They are shown in **Figures 17 and 18**.

Step

All Steps

Setup Number

Group 1 [Setups 1, 2]

Main Settings

Timelag (s)

1

Number of Singular Values to plot

1

Data Channels

Channels

☐ Vx
☐ Vy
☐ Vz
☒ Ax
☒ Ay
☒ Az

Removed Sensor IDs

☐ Use reference sensor(s)

Figure 17: SSI Parameters, Part I

Signal Processing

☐ Filter

☐ Use Equalization

☐ Decimate

☐ Perform Cleanup

Lowest Model Order

2

Highest Model Order

50

Clustering Threshold Factor

0.2

Figure 18: SSI Parameters, Part II

Results

The results section for multi-setup cases is more complex than for single-setup cases. In the current version of the Dashboard, **manual mode merging** is required. To merge modes, users must select and label the corresponding modes in each setup. The number of selected modes should be the same across setups.

Alternatively, users can label the desired modes in just one setup and click the **“Apply Selections for All Tabs”** button. This feature automatically searches for modes with similar frequencies in the other setups and applies the labels accordingly.

The selected modes for Setup 1 and Setup 2 are shown in **Figures 19 and 20**.

Modes Threshold

Auto-Select Frequency Delta

0.1

Reset Analysis

☐ Only Show Selected Modes

Setup 1

Setup 2

Merged

Detected Modes

Show

10

entries

Apply Selections to All Tabs

Export Selected

Save Shapes

Merge Modes

Search:

☐	Number of Poles	Frequency (Hz)	Frequency std (Hz)	Damping (%)	Damping std (%)	Polar Chart	Mode Shape
☒	22	4.415	0.001	4.543	0.050	Display	Plot
☒	17	5.883	0.015	5.836	0.469	Display	Plot
☒	18	9.245	0.036	4.212	1.299	Display	Plot
☐	13	15.700	0.014	4.611	0.184	Display	Plot

Figure 19: SSI Result – Setup 1

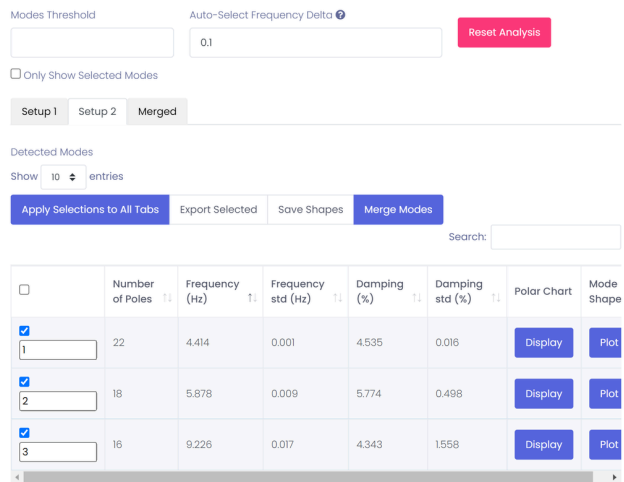


Figure 20: SSI Result – Setup 2

Once the desired modes are labeled, users can click the **“Merge Modes”** button to combine them across setups. The merged results tab is shown in **Figure 21**, and the merged modes are displayed in **Figures 22, 23, and 24**.

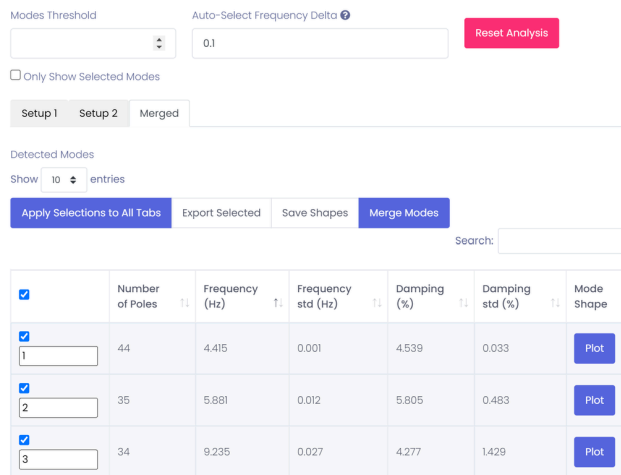


Figure 21: SSI Result – Merged Modes

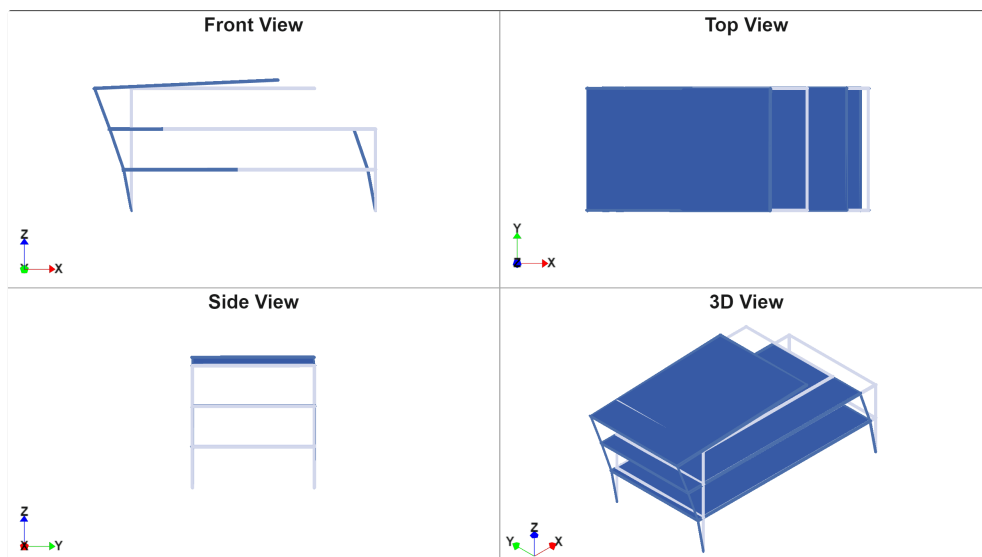


Figure 22: Merged Mode at 4.415 Hz

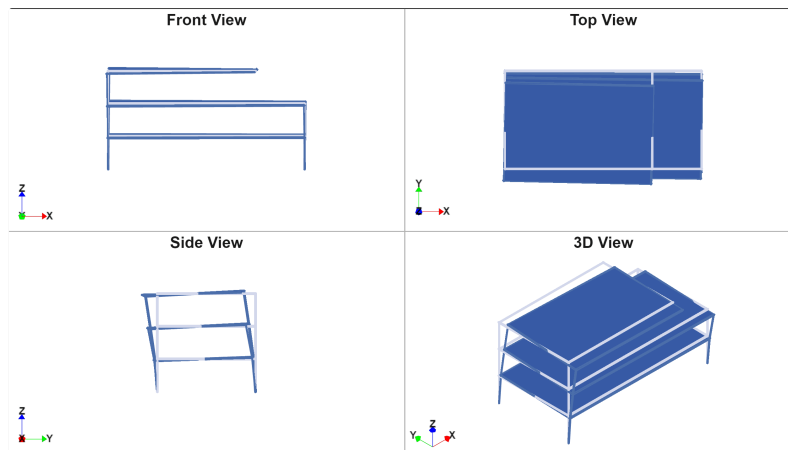


Figure 23: Merged Mode at 5.881 Hz

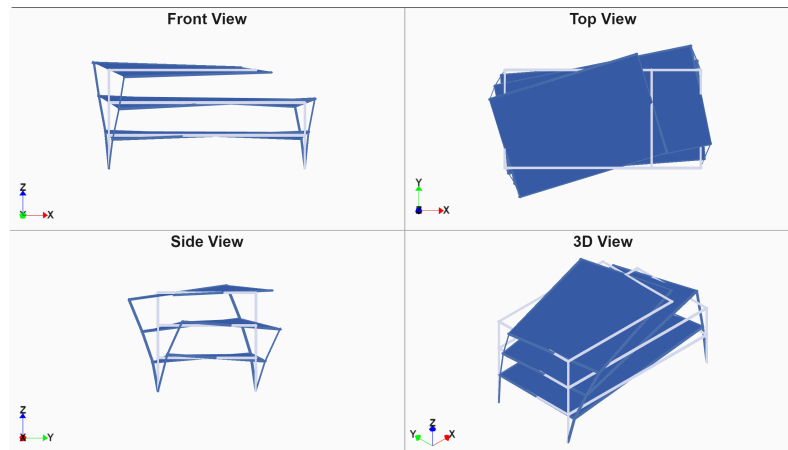


Figure 24: Merged Mode at 9.235 Hz

Users can export mode shape values and mode shape animations once the analysis is complete.

FAQ: Modal Analysis SSI

How should I choose the time lag for SSI analysis?

In stochastic subspace identification (SSI), the time lag is a key parameter that controls the length of the correlation function, which captures the temporal relationships between different time series data. This correlation function is essential for extracting modal parameters such as natural frequencies, damping ratios, and mode shapes from ambient vibration or other structural response data.

The time lag determines how far back in time the correlation is computed, directly affecting the amount of dynamic information included in the analysis. As a general guideline, the time lag should be set to **3 to 10 times the period of the target mode**. This range ensures the correlation function covers multiple oscillation cycles of the mode, which is critical for accurately identifying its dynamic characteristics.

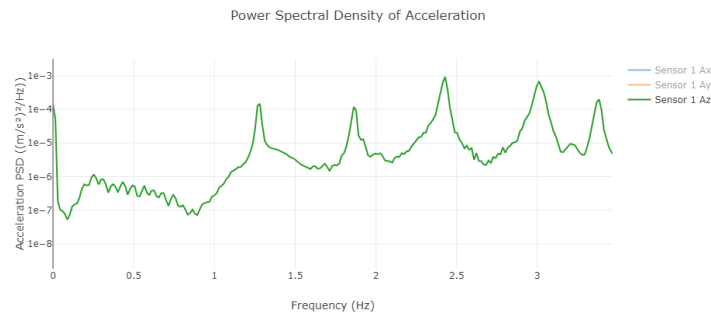


Figure 25: Z-axis Acceleration PSD of Reference Sensor

Example of Time Lag

To illustrate, consider a demo project targeting two modes at 1.3 Hz and 1.9 Hz. The power spectral density (PSD) of the reference sensor is shown in Figure 25. For analyzing these modes, a time lag of 4 seconds is selected. This time lag covers more than 5 periods of the 1.3 Hz mode and over 7 periods of the 1.9 Hz mode, providing a strong basis for accurate correlation.

The resulting correlation function, calculated with a 4-second time lag, is shown in Figure 26. A bandpass filter was applied during this process to reduce the influence of higher-frequency modes. The choice of filters is discussed in a separate section.

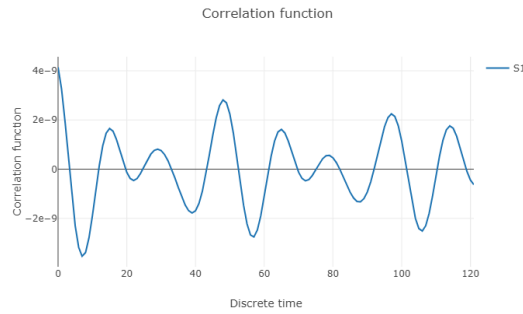


Figure 26: Correlation Function with Appropriate Time Lag

Impacts of Incorrect Time Lag

If the time lag is too short—as shown in Figure 27, where a 0.5-second time lag was used—the correlation function may not capture enough information, leading to reduced SSI analysis quality. In this case, the correlation function covers less than one period of the 1.9 Hz mode and about half a period of the 1.3 Hz mode.

Conversely, if the time lag is too long—as illustrated in Figure 28 with a 20-second time lag—the correlation function may include excessive noise and unnecessarily increase computational demands.

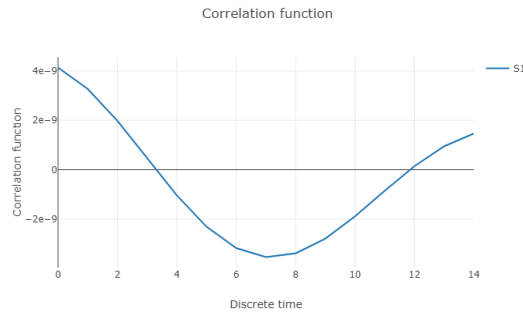


Figure 27: Correlation Function Using an Insufficiently Short Time Lag

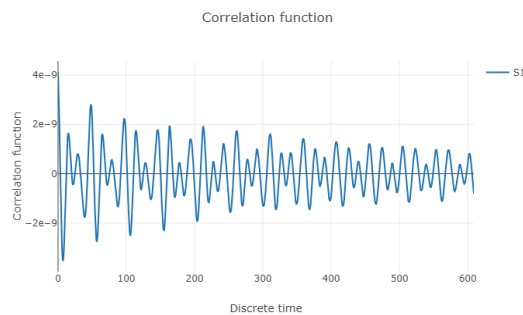


Figure 28: Correlation Function with an Excessively Long Time Lag

The number of timelag steps—calculated by multiplying the time lag by the sampling frequency—should ideally be between **80 and 150 steps**. This range is optimal, providing enough data for accurate analysis while keeping the computational load manageable.

The selection of the time lag should be made in conjunction with the decimation factor, which adjusts the sampling frequency. Together, these parameters help ensure the correlation function is both informative and efficient. For example, if the original sampling frequency is much higher than the target mode frequency, applying a decimation factor reduces the sampling frequency and brings the number of time lag steps into the optimal range without sacrificing analysis quality. The choice of decimation factor, which directly affects the final sampling frequency and the number of time lag steps, will be discussed in the next section.

How does decimation work?

Decimation is a technique used to reduce the sampling frequency of a signal while preserving the important content within the target frequency range. It is especially useful when the frequency of the target mode is much lower than the original sampling frequency, enabling more efficient data processing without sacrificing accuracy.

Decimation works by systematically down-sampling the data, which involves filtering out unnecessary high-frequency components that are irrelevant to the analysis. The goal is to balance reducing the sampling frequency enough to improve computational efficiency while maintaining the fidelity of the target mode's frequency content.

The choice of decimation factor is closely linked to the selection of time lag, as both parameters influence the number of time lag steps—an important factor in stochastic subspace identification (SSI). The optimal number of time lag steps typically ranges between 80 and 150, providing a robust foundation for reliable modal analysis.

Example of Decimation Factor Selection

Let's revisit the example from the time lag section. For analyzing the 1.3 Hz and 1.9 Hz modes, a 4-second time lag was chosen. The original sampling frequency for this test is 122 Hz, which, while providing high-resolution data, can be excessive for the lower frequency range of interest.

To bring the number of time lag steps into the optimal range, a decimation factor can be applied. In this case, selecting a decimation factor of 4 reduces the sampling frequency to 30.5 Hz. With this adjustment, the 4-second time lag corresponds to $4 \text{ s} \times 30.5 \text{ Hz} = 122$ time lag steps, which falls comfortably within the desired range. The correlation function using this configuration is shown in **Figure 26**, demonstrating that the decimated data still accurately captures the essential characteristics of the target modes.

Impacts of Incorrect Decimation Factor Selection

Careful selection of the decimation factor is essential. If the decimation factor is set too high, the resulting reduction in sampling frequency can cause aliasing—where higher-frequency components are misinterpreted as lower frequencies. This leads to degraded data quality, evident in the SSI analysis as noise and distortions in the correlation function that obscure the true modal properties. **Figure 29** illustrates this scenario, where a 4-second time lag and a decimation factor of 20 were used. Although the general shape of the correlation function resembles that in **Figure 26**, its clarity and quality are significantly reduced, which may result in inaccurate mode identification.

Conversely, if the decimation factor is too low or no decimation is applied, the computational burden increases dramatically. Without decimation, the data remains at the original high sampling frequency, producing an excessive number of time lag steps. For example, in **Figure 30**, a 4-second time lag with no decimation leads to 488 time lag steps ($4 \text{ s} \times 122 \text{ Hz}$), far exceeding the optimal range. While the correlation function does not visibly improve compared to **Figure 26**, the computational cost rises steeply because the singular value decomposition (SVD) used in SSI has a time complexity of $O(n^3)$. This exponential increase in processing time can make analysis impractical, especially for large datasets or real-time applications.

In practice, determining the optimal decimation factor involves both theoretical guidance and empirical testing. It is important to consider the structural characteristics, the frequency range of interest, and the capabilities of the data

acquisition system. By carefully balancing the time lag, decimation factor, and sampling frequency, one can achieve efficient and accurate SSI analysis, yielding valuable insights into the structural dynamics of the system under investigation.

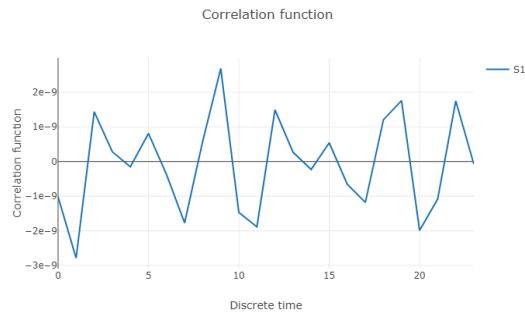


Figure 29: Correlation Function with Excessive Decimation

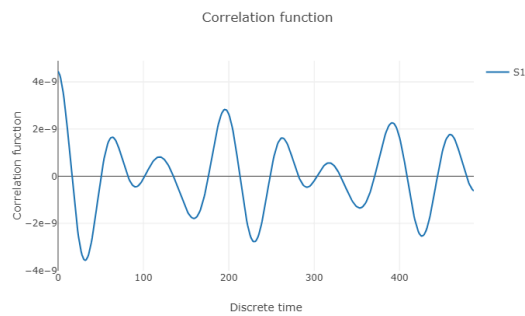


Figure 30: Correlation Function without Decimation

What is data cleanup? Should I use it?

Data cleanup detects and removes unintended high-amplitude spikes in sensor data, which can occur due to accidental impacts—such as a sensor being kicked during a test. This feature is valuable for maintaining data quality, especially in applications like Structural Health Monitoring (SHM), where such unexpected signals can interfere with analysis methods like Stochastic Subspace Identification (SSI) and Frequency Domain Decomposition (FDD).

How It Works

- Segmenting the Signal:** The data cleanup function divides the signal into small segments of equal length, defined by the parameter **“Cleanup Granularity (s).”** For each segment, the Root Mean Square (RMS) value is calculated to represent its overall energy.
- Thresholding for Spikes:** Using the median RMS value across all segments as a baseline, the function applies a **“Cleanup Threshold”** to identify outliers. Segments with RMS values exceeding this threshold are considered to contain unintended spikes—likely caused by impacts or noise—and are removed from the data.

When to Use Data Cleanup

While data cleanup can enhance signal quality by removing unintended spikes, it must be used with caution to avoid discarding meaningful structural responses. It's generally recommended to **begin your analysis without cleanup**. If the results are not as expected—such as distorted mode shapes or excessive curvature changes—then applying cleanup may help improve the outcome by eliminating anomalies.

If the spikes reflect genuine structural behavior (e.g., due to ambient excitation), using cleanup may inadvertently remove valid data, potentially lowering the signal-to-noise ratio and degrading analysis quality.

There is no universal rule for applying data cleanup—it is a **judgment call** that depends on understanding the nature of the spike and the testing conditions. Here are some guidelines to help you decide:

- **Look for consistent patterns:** If the spike fits the general behavior of the signal or aligns with expected vibration modes, it may be a legitimate structural response. External impacts usually appear as sudden, isolated anomalies with no consistent pattern before or after.
- **Check across sensors:** True structural responses typically appear across multiple sensors. If the spike is isolated to a single sensor, it may be due to local interference or a sensor-specific issue.
- **Evaluate the impact on mode shapes:** Run the analysis with and without the affected segment. If clean mode shapes still appear, the spike may be harmless. If mode shapes degrade or become inconsistent, cleanup could improve the result.
- **Consider test conditions:** Was the environment controlled? Were sensors securely placed? Were there people or equipment moving nearby? Cleanup is more useful in environments where external disturbances are likely.

Ultimately, the decision to use data cleanup depends on the **quality of your data**, the **testing context**, and whether the spike **compromises the integrity of the results**.

Note: The goal is to isolate and remove only the most obvious spikes while preserving the integrity of the overall signal. The RMS-based threshold targets extreme spikes that clearly stand out. Be careful not to set the threshold too low to avoid removing valid data.

Geophone Equalization

Geophone equalization compensates for the natural roll-off in the low-frequency response typical of geophone sensors. This feature extends the effective frequency range for analysis, making it especially valuable for applications like ambient vibration testing where accurate low-frequency response is crucial.

Understanding Geophone Roll-off at Low Frequencies

Geophones are velocity sensors with a limited response at low frequencies, causing signal attenuation—or “roll-off”—below a certain frequency range. This behavior results from the geophone's internal spring-mass-damping system, which is optimized for specific frequencies and affects how lower-frequency signals are captured. Without equalization, these low-frequency signals may be underrepresented, potentially impacting analyses that rely on accurate low-frequency data.

How Equalization Works

The equalization function corrects the low-frequency response of each geophone based on key parameters:

- **Natural Frequency:** The frequency at which the geophone is most responsive. Equalization boosts signals below this frequency to compensate for roll-off.

- **Damping:** This defines how quickly the geophone returns to rest and affects how frequencies near the natural frequency are attenuated or amplified.
- **Sensitivity:** The geophone's baseline response level, which determines the gain applied during equalization to balance the output across the frequency range.

Sensequake's sensors carefully record these parameters and store them in the sensors' header files. When geophone equalization is enabled, the algorithm automatically reads and applies these parameters during the equalization process.

When to Use Equalization

Equalization is recommended for velocity channels when accurate low-frequency data (below 5 Hz) is critical for analysis. This is especially important in applications requiring full-spectrum response, such as structural health monitoring of large structures, seismic analysis, or any scenario where low-frequency content offers valuable insights into system behavior.